

Software Requirements Specification (SRS)

Project Name: OpenScholar

Version: 1.0

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Version History

Version	Date	Description	Author
1.0	23 Feb 2026	Initial SRS Draft	Md. Shahriar Hossain Jihad

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the requirements for **OpenScholar**, an open-access academic research publishing and sharing platform.

This document provides a detailed description of the system's functional and non-functional requirements and serves as a reference for developers, testers, project supervisors, and university stakeholders.

Unlike traditional academic portals that focus only on functionality, this project emphasizes:

- Visual communication
- User-centered design
- Interactive interface design
- Typography and layout systems
- Data visualization
- Multimedia integration

The core objective is to design a visually engaging, minimalistic, and user-friendly academic platform that enhances research discoverability through advanced UI/UX principles.

1.2 Background and Problem Statement

Most academic research platforms suffer from:

- Research papers locked behind expensive paywalls.
- Access requires institutional or business email credentials.
- Student research lacks visibility
- no centralized platform in Bangladesh.
- Outdated interface design
- Poor typography and readability
- Cluttered dashboards
- Complex navigation
- Lack of visual hierarchy
- Limited multimedia integration
- Poor mobile responsiveness

As a result, users experience:

- Cognitive overload
- Low engagement
- Reduced usability
- Poor reading experience

There is a need for a **modern, graphics-driven, interaction-focused academic platform** that improves the research experience through effective UI/UX design.

1.3 Project Objectives

The primary objectives of OpenScholar are:

1. To provide a free and open-access research repository.
2. To enable students to publish thesis and research papers easily.
3. To increase research visibility within universities.
4. To support multilingual interface (Bangla & English).
5. To implement a secure, moderated publication workflow.
6. To create a scalable system that can expand nationally and globally.

1.4 Proposed Solution

OpenScholar will be a modern open source web application with a scalable backend architecture.

The platform will provide:

- Student profile creation and authentication
- Research paper upload and metadata management
- Admin approval workflow
- Public search and filtering system
- Free PDF reading and downloading
- Analytics dashboard
- Multilingual support
- Role-based access control

The system will follow secure authentication protocols and open-access policies.

1.5 Scope

In Scope

- Web-based academic repository
- Student account registration
- Research paper submission system
- Admin moderation system
- Search and filter functionality
- Download and citation export
- Dashboard and analytics
- Language switch feature

Out of Scope (Initial Phase)

- DOI generation
- International indexing integration
- Peer-reviewed journal management
- Paid subscription system

1.6 Intended Audience and Use

This document is intended for:

- Software Developers
- QA/Test Engineers
- University Administration
- Project Supervisors
- Academic Review Committee

It will be used as a contractual and technical reference for system development and validation.

1.7 Target Users

1. Undergraduate Students
2. Graduate Students
3. Research Scholars
4. Faculty Members
5. University Administrators
6. Public Researchers

1.8 Definitions, Acronyms, Abbreviations

Term	Definition
SRS	Software Requirements Specification
Admin	System Administrator
DOI	Digital Object Identifier
RBAC	Role-Based Access Control
PDF	Portable Document Format
API	Application Programming Interface

1.9 References

1. Software Requirements Specification Guide
 2. Pressman, R., & Maxim, B., *Software Engineering: A Practitioner's Approach*
 3. IEEE Std 830-1998
 4. ISO/IEC/IEEE 29148:2018
 5. Project PRD (OpenScholar)
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2. Overall Description

2.1 Product Perspective

OpenScholar is a standalone web-based platform.

It acts as:

- A standalone academic platform
- Act as a centralized academic repository.
- Integrate with university email systems (optional).
- Operate on cloud infrastructure.
- A scalable SaaS-ready system

Future integrations:

- University systems
 - DOI providers
 - Plagiarism detection APIs
 - National research databases
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2.2 Product Functions (High-Level Features)

- User Registration & Authentication
 - Profile management
 - Role-Based Access Control
 - Research Paper Submission
 - Admin moderation
 - Search & Filtering System
 - Multilingual Support
 - Paper viewing
 - Analytics Dashboard
 - Download & Citation Export
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2.3 User Classes and Characteristics

User Class	Description	Technical Expertise
Student	Uploads research	Basic–Moderate
Researcher	Reads papers & Verifies publications	Moderate
Admin	Manages system & Moderates content	Advanced
Public User	Reads/downloads papers	Basic

2.4 Operating Environment

- Web Application
 - Supported Browsers: Chrome, Firefox, Edge
 - Backend Server (Cloud-based - Linux)
 - Database Server (Relational DB)
 - Operating System: Linux-based server
 - Internet connection required
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2.5 Design and Implementation Constraints

- Secure authentication system required.
 - Must support large file uploads
 - Must ensure secure authentication
 - Must be scalable
 - Must follow open-access policy
 - Must comply with university publication rules.
 - Must ensure data privacy compliance.
 - Large file upload support ($\geq 50\text{MB}$).
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2.6 Assumptions and Dependencies

Assumptions

- Universities will verify student identity.
- Universities will cooperate

- Students will upload original work.
- Internet connectivity is available.

Dependencies

- Cloud hosting provider.
 - Email service
 - Storage service
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3. Specific Requirements

3.1 Functional Requirements

3.1.1 User Registration

Description: Users can create an account.

Priority: Must-have

Stimulus: User submits registration form.

Response: System validates and creates account.

Functional Requirements:

- FR-1.1: The system shall provide a registration form.
 - FR-1.1: The system shall support multiple types of user onboarding (Student, Researcher, Professional, Normal user)
 - FR-1.2: The system shall require email verification.
 - FR-1.3: The system shall allow role selection (Student/Admin).
 - FR-1.4: The system shall prevent duplicate email registration.
 - FR-1.5: The system shall store user credentials securely.
 - FR-1.6: The system shall maintain user sessions.
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3.1.2 User Login

Description: Authenticated access to dashboard.

Priority: Must-have

- FR-2.1: The system shall provide login form.
 - FR-2.2: The system shall validate credentials.
 - FR-2.3: The system shall redirect authenticated users to the dashboard.
 - FR-2.4: The system shall show an error message for invalid login.
 - FR-2.5: The system shall support password reset.
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3.1.3 User Profile

- FR-3.1: The system shall allow profile editing.
 - FR-3.2: The system shall store academic information.
 - FR-3.3: The system shall display user publications.
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3.1.4 Research Paper Submission

Description: Student uploads research paper.

Priority: Must-have

- FR-4.1: The system shall allow PDF upload.
 - FR-4.2: The system shall require metadata (title, abstract, and keywords).
 - FR-4.3: The system shall allow department selection.
 - FR-4.4: The system shall mark the submission as "Pending Approval."
 - FR-4.5: The system shall support version updates.
 - FR-4.6: The system shall allow draft saving.
 - FR-4.7: The system shall store paper author and co-author info
 - FR-4.8: The system shall store paper's title, keyword, metadata, short description, and category info for each version
 - FR-4.9: The System shall support external authors
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3.1.5 Admin Approval

Description: Admin reviews submissions.

Priority: Must-have

- FR-5.1: The system shall allow admin to approve/reject.
 - FR-5.2: The system shall notify students of the decision.
 - FR-5.3: The system shall publish approved papers publicly.
 - FR-5.4: The system shall log moderation history.
 - FR-5.5: Admin shall manage users (view, suspend, activate)
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3.1.6 Search and Discovery

Description: Users search research papers.

Priority: Must-have

- FR-6.1: The system shall support keyword search.
 - FR-6.2: The system shall filter by research category and subcategory (Science, physics, chemistry, AI, computer vision, LLM, cybersecurity, etc.).
 - FR-6.3: The system shall filter by university.
 - FR-6.4: The system shall filter by department.
 - FR-6.5: The system shall filter by publication year.
 - FR-6.6: The system shall display search results within 2 seconds.
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3.1.7 Download & Reading

- FR-7.1: The system shall display paper details.
 - FR-7.2: The system shall provide a PDF viewer.
 - FR-7.3: The system shall allow free download.
 - FR-7.4: The system shall record download statistics.
 - FR-7.5: The system shall generate citation formats.
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3.1.8 Multilingual Support

- FR-8.1: The system shall provide a language switch option.
 - FR-8.2: The system shall support English and Bangla.
 - FR-8.3: The system shall save user language preference.
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3.1.9 Analytics Dashboard

- FR-9.1: The system shall display total publications.
- FR-9.2: The system shall track views.
- FR-9.3: The system shall display the most viewed papers.
- FR-9.4: The system shall display university-wise statistics.
- FR-9.5: The system shall track downloads.
- FR-9.6: The system shall display statistics.

3.1.10 User Features

- FR-10.1: The system shall allow reaction (Like, Love, Haha, Intelligent, etc.) to any published paper
 - FR-10.2: The system shall allow commenting and replying to comments by any logged-in or non-logged-in user.
 - FR-10.3: The system shall allow social media sharing of any published paper
 - FR-10.4: Users shall be able to follow/unfollow other users
 - FR-10.5: Users shall see suggested researchers
 - FR-10.6: Users shall be able to save (bookmark) papers
 - FR-10.7: Users shall be able to view saved papers
 - FR-10.8: The system shall display related papers based on category and keywords
 - FR-10.9: The system shall display global platform statistics (papers, users, downloads)
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3.2 Non-Functional Requirements

3.2.1 Performance

- NFR-1: 95% of pages shall load within 3 seconds.
 - NFR-2: System shall support 10,000 concurrent users.
 - NFR-3: Search results shall appear within 2 seconds.
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3.2.2 Security

- NFR-4: Passwords shall be encrypted.
 - NFR-5: System shall implement RBAC.
 - NFR-6: System shall prevent SQL injection.
 - NFR-7: Data transmission shall use HTTPS.
 - NFR-8: Admin access shall require multi-factor authentication (future).
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3.2.3 Usability

- NFR-9: A new user shall complete registration within 3 minutes.
- NFR-10: Interface shall be mobile responsive.
- NFR-11: Navigation shall require no more than 3 clicks to reach any major feature.
- NFR-12: UI must be responsive

3.2.4 Reliability

- NFR-13: System uptime shall be 99%.
 - NFR-14: Daily automated backup required.
 - NFR-15: Data recovery shall be possible within 4 hours.
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3.2.5 Scalability

- NFR-16: System must support multi-university expansion
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3.2.6 Maintainability

- NFR-17: Code shall follow modular architecture.
 - NFR-18: API documentation shall be maintained.
 - NFR-19: System shall support version upgrades without downtime.
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3.2.7 Portability

- NFR-20: System shall be deployable on major cloud providers.
 - NFR-21: System shall support modern browsers.
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3.3 External Interface Requirements

3.3.1 User Interfaces

- Dashboard interface.
- Submission form.
- Admin moderation panel.
- Search results page.

- Language switch option.
 - PDF viewer interface.
 - Focus Mode
 - Dark mode support
 - Reaction, comments and reply to comment
 - Share to social Media
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3.3.2 Hardware Interfaces

- Compatible with desktop, laptop, tablet, and mobile devices.
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3.3.3 Software Interfaces

- Email service (SMTP).
 - Database server (Relational DB).
 - Cloud storage system.
 - Future plagiarism API integration.
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3.3.4 Communications Interfaces

- HTTPS protocol.
 - RESTful API (future mobile app integration).
 - JSON data exchange format.
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4. Supporting Information

4.1 Requirements Traceability Matrix (RTM)

4.1.1 Functional Requirements Traceability Matrix

Req ID	Requirement Description	Module	Use Case	Test Case ID
FR-1.1	Registration form	Auth	User Onboarding	TC-AUTH-01
FR-1.2	Email verification	Auth	Verify Email	TC-AUTH-02
FR-1.3	Role selection	Auth	Assign Role	TC-AUTH-03
FR-1.4	Prevent duplicate email	Auth	Validate Email	TC-AUTH-04
FR-1.5	Secure credential storage	Auth	Store User	TC-AUTH-05
FR-1.6	Maintain session	Auth	Login Session	TC-AUTH-06

Req ID	Requirement Description	Module	Use Case	Test Case ID
FR-2.1	Login form	Auth	Login User	TC-AUTH-07
FR-2.2	Validate credentials	Auth	Authenticate	TC-AUTH-08
FR-2.3	Redirect to dashboard	Auth	Login Flow	TC-AUTH-09

FR-2.4	Error on invalid login	Auth	Error Handling	TC-AUTH-10
FR-2.5	Password reset	Auth	Reset Password	TC-AUTH-11
FR-3.1	Profile editing	User	Update Profile	TC-USER-01
FR-3.2	Store academic info	User	Save Info	TC-USER-02
FR-3.3	Show publications	User	View Papers	TC-USER-03

FR-4.1	Upload PDF	Paper	Submit Paper	TC-PAPER-01
FR-4.2	Add metadata	Paper	Add Details	TC-PAPER-02
FR-4.3	Select department	Paper	Assign Dept	TC-PAPER-03
FR-4.4	Set pending status	Paper	Submission Flow	TC-PAPER-04
FR-4.5	Version update	Paper	Update Paper	TC-PAPER-05
FR-4.6	Save draft	Paper	Draft Save	TC-PAPER-06

FR-5.1	Approve/reject paper	Admin	Moderation	TC-ADMIN-01
FR-5.2	Notify user	Admin	Notification	TC-ADMIN-02
FR-5.3	Publish paper	Admin	Publish Flow	TC-ADMIN-03
FR-5.4	Log moderation	Admin	Audit Log	TC-ADMIN-04

FR-6.1	Keyword search	Search	Search Query	TC-SEARCH-01
FR-6.2	Filter by category	Search	Filter Category	TC-SEARCH-02
FR-6.3	Filter by university	Search	Filter Univ	TC-SEARCH-03
FR-6.4	Filter by department	Search	Filter Dept	TC-SEARCH-04
FR-6.5	Filter by year	Search	Filter Year	TC-SEARCH-05
FR-6.6	Response within 2 sec	Search	Performance	TC-SEARCH-06

FR-7.1	Show paper details	Reading	View Paper	TC-READ-01
FR-7.2	PDF viewer	Reading	Read Paper	TC-READ-02
FR-7.3	Free download	Reading	Download	TC-READ-03
FR-7.4	Track downloads	Analytics	Log Download	TC-ANALYTICS-01
FR-7.5	Citation export	Reading	Export Citation	TC-READ-04

FR-8.1	Language switch	UI	Change Language	TC-UI-01
FR-8.2	English/Bangla support	UI	Localization	TC-UI-02
FR-8.3	Save preference	UI	Persist Lang	TC-UI-03

FR-9.1	Show total publications	Analytics	Dashboard	TC-ANALYTICS-02
FR-9.2	Track views	Analytics	View Count	TC-ANALYTICS-03
FR-9.3	Most viewed papers	Analytics	Ranking	TC-ANALYTICS-04
FR-9.4	University stats	Analytics	Stats	TC-ANALYTICS-05
FR-9.5	Track downloads	Analytics	Metrics	TC-ANALYTICS-06
FR-9.6	Show statistics	Analytics	Dashboard	TC-ANALYTICS-07

4.1.2 Non-Functional Requirements Traceability Matrix

2.1 Performance

Req ID	Requirement	Test Case
NFR-1	Page load < 3 sec	TC-PERF-01
NFR-2	10k users support	TC-PERF-02
NFR-3	Search < 2 sec	TC-PERF-03

2.2 Security

Req ID	Requirement	Test Case
NFR-4	Password encryption	TC-SEC-01

NFR-5	RBAC	TC-SEC-02
NFR-6	SQL injection prevention	TC-SEC-03
NFR-7	HTTPS	TC-SEC-04
NFR-8	MFA (future)	TC-SEC-05

2.3 Usability

Req ID	Requirement	Test Case
NFR-9	Registration < 3 min	TC-UX-01
NFR-10	Mobile responsive	TC-UX-02
NFR-11	≤3 clicks navigation	TC-UX-03
NFR-12	Responsive UI	TC-UX-04

2.4 Reliability

Req ID	Requirement	Test Case
NFR-13	99% uptime	TC-REL-01
NFR-14	Daily backup	TC-REL-02
NFR-15	Recovery within 4h	TC-REL-03

2.5 Scalability

Req ID	Requirement	Test Case
NFR-16	Multi-university support	TC-SCALE-01

2.6 Maintainability

Req ID	Requirement	Test Case
NFR-17	Modular architecture	TC-MAIN-01
NFR-18	API documentation	TC-MAIN-02

NFR-19	Version upgrade support	TC-MAIN-03
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2.7 Portability

Req ID	Requirement	Test Case
NFR-20	Cloud deployable	TC-PORT-01
NFR-21	Browser support	TC-PORT-02

4.1.3 Interface Requirements Traceability Matrix

3.1 UI Interfaces

Feature	Test Case
Dashboard	TC-UI-05
Submission form	TC-UI-06
Admin panel	TC-UI-07
Search page	TC-UI-08
PDF viewer	TC-UI-09
Focus mode	TC-UI-10
Dark mode	TC-UI-11

3.2 Software Interfaces

Interface	Test Case
Email SMTP	TC-INT-01
Database	TC-INT-02

Cloud Storage	TC-INT-03
API Integration	TC-INT-04

3.3 Communication Interfaces

Interface	Test Case
HTTPS	TC-COM-01
REST API	TC-COM-02
JSON format	TC-COM-03

4.2 Analysis Models (To Be Attached)

- Use Case Diagram
- ER Diagram
- Activity Diagram
- Data Flow Diagram (DFD)

4.3 Approval and Sign-off

Role	Name	Signature	Date
Project Owner	Md. Shahriar Hossain Jihad		
Supervisor			
Developer	Md. Shahriar Hossain Jihad		

Conclusion

This SRS defines the complete requirement specification of OpenScholar following standard SRS structure and principles. All requirements are written in clear, testable, and unambiguous format using “shall” statements as recommended in SRS best practices